Substance Use & School Violence Inference Analysis

RESEARCH QUESTION

Does the frequency of adolescent substance use (smoking & alcohol) significantly predict the risk of school violence (fighting & carrying weapons)?

METHOD

- Ordinary Least Squares (OLS) Multiple Linear Regression
- Violence = β0+ β1·Cig + β2·Alc +β3·Age + β4·Sex + ε

RESULTS

Predictor	Coef (β)	p-value
Smoking Days	0.0260	<0.001
Drinking Days	0.0849	<0.001
Gender (Male)	0.5093	<0.001
Age	-0.0736	<0.001

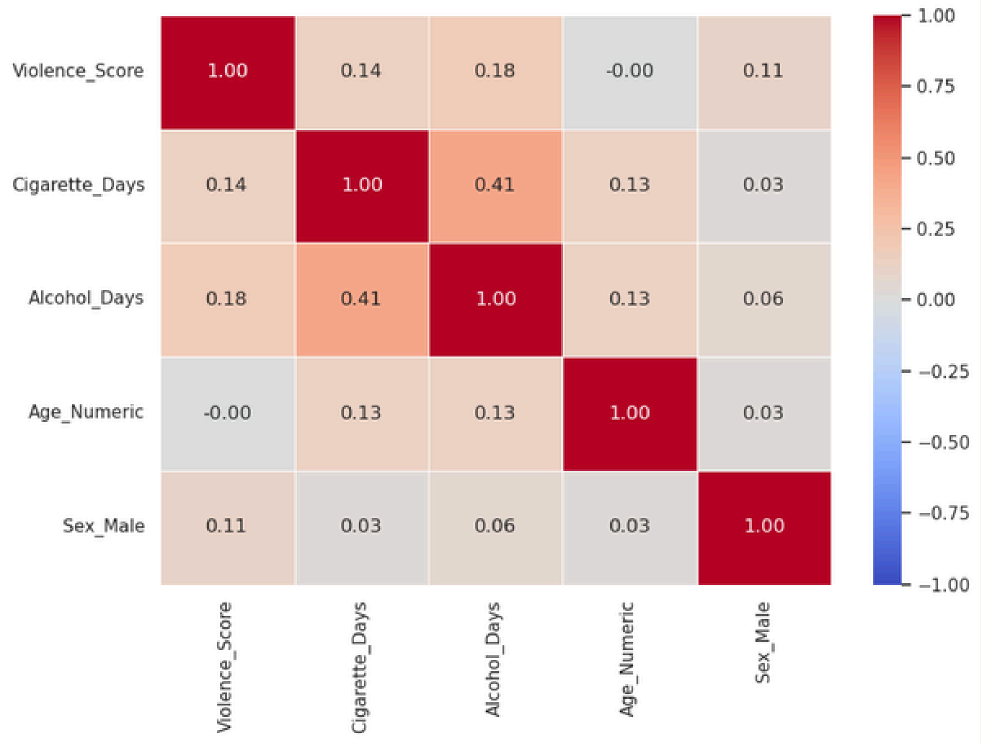
R-squared: 0.050 | F-statistic: 156.8 (p < 0.001)

DATA & VARIABLES

- Dependent (Y): School Violence Index (Fights + Weapons)
- Independent (X): Cigarette Smoking Days,Alcohol Drinking Days
- Controls: Age, Gender

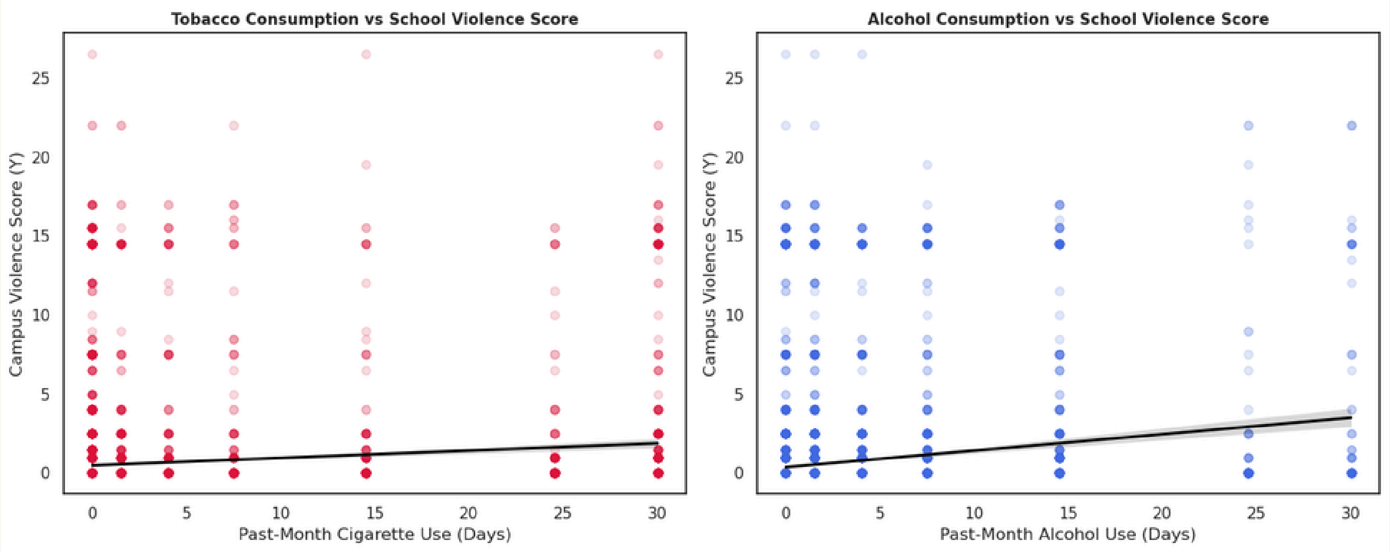
EDA

Figure 1: Correlation Matrix Heatmap



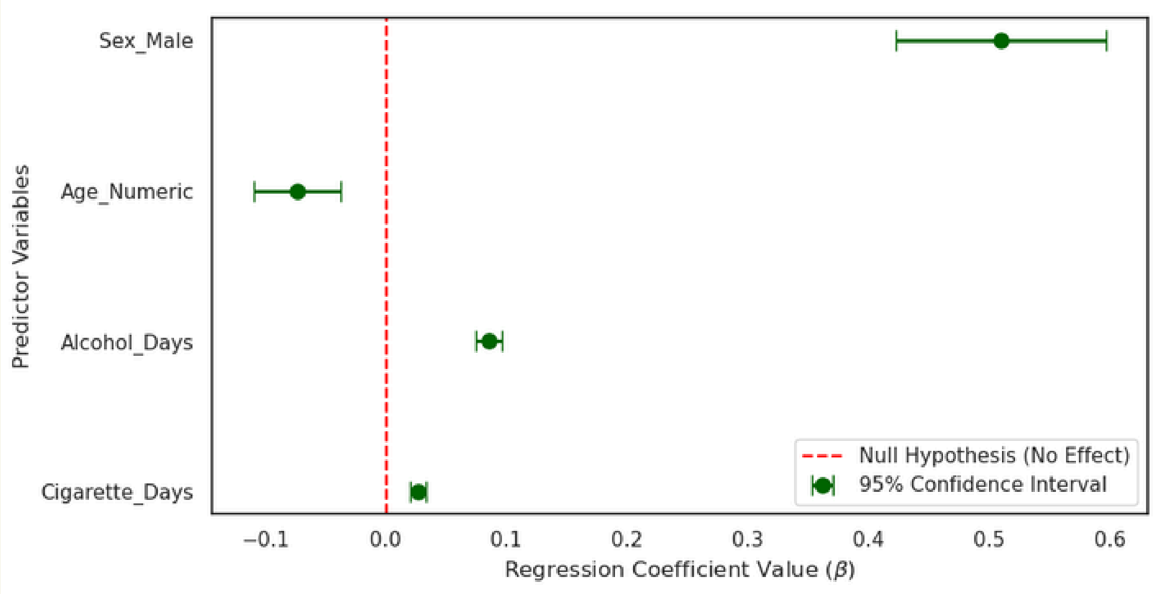
As shown in the heatmap, based on a massive sample size of N= 11,904, we see a strong positive correlation cluster between substance use and school violence risk.

Figure 2: Linear Regression Trends for Substance Abuse Predictors



The linear trend clearly demonstrates that school violence risk escalates with drinking frequency, heavily driven by its top-impact coefficient of (β) = 0.0849."

Figure 3: Predictor Coefficients with 95% Confidence Intervals



The coefficient plot confirms that smoking, drinking, and male status are significant positive predictors, with all 95% confidence intervals above zero and p < 0.001.

(Conclusion)

The null hypothesis is successfully rejected. Empirical evidence confirms that the frequency of adolescent substance use is a robust, positive predictor of school violence risks.